

SDITO LAB

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Assignment 07

Problem Statement : Hosting a website on EC2

Aim

To launch an Amazon EC2 instance, configure a web server, and host a static website using terminal-based configuration.

Requirements

- AWS Account
- Key pair (.pem file)
- HTML file (index.html)
- Internet connection
- Bitwise SSH Client

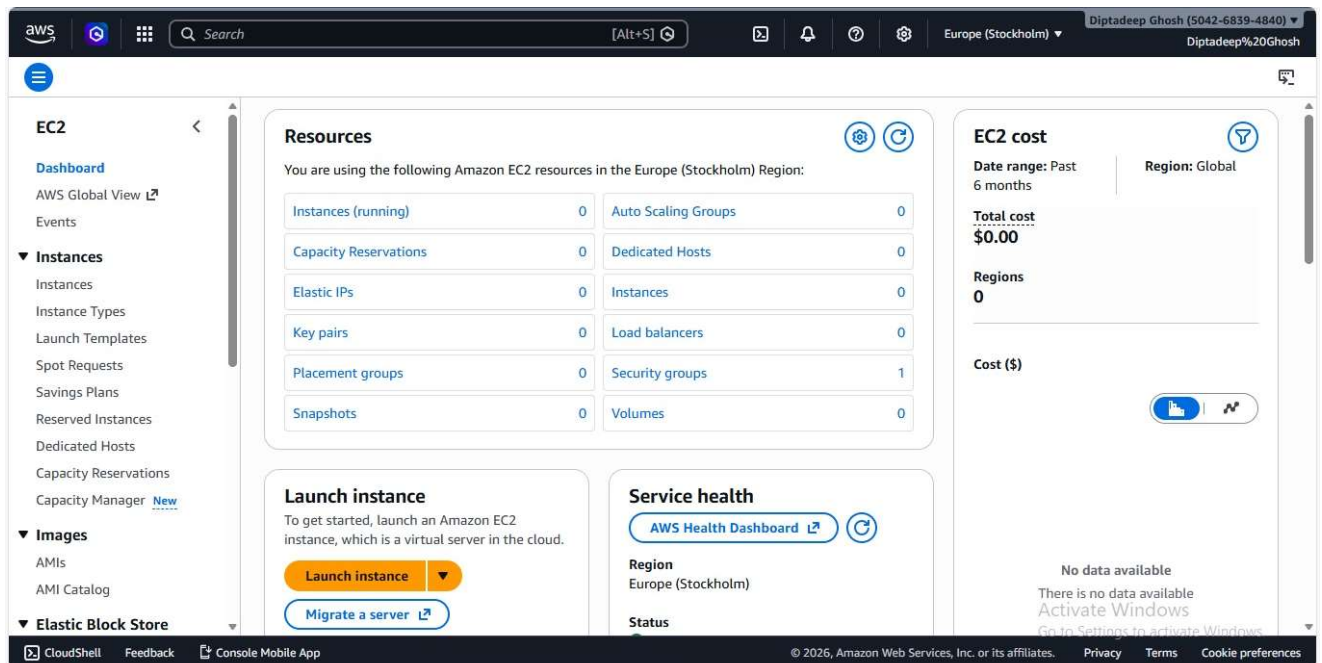
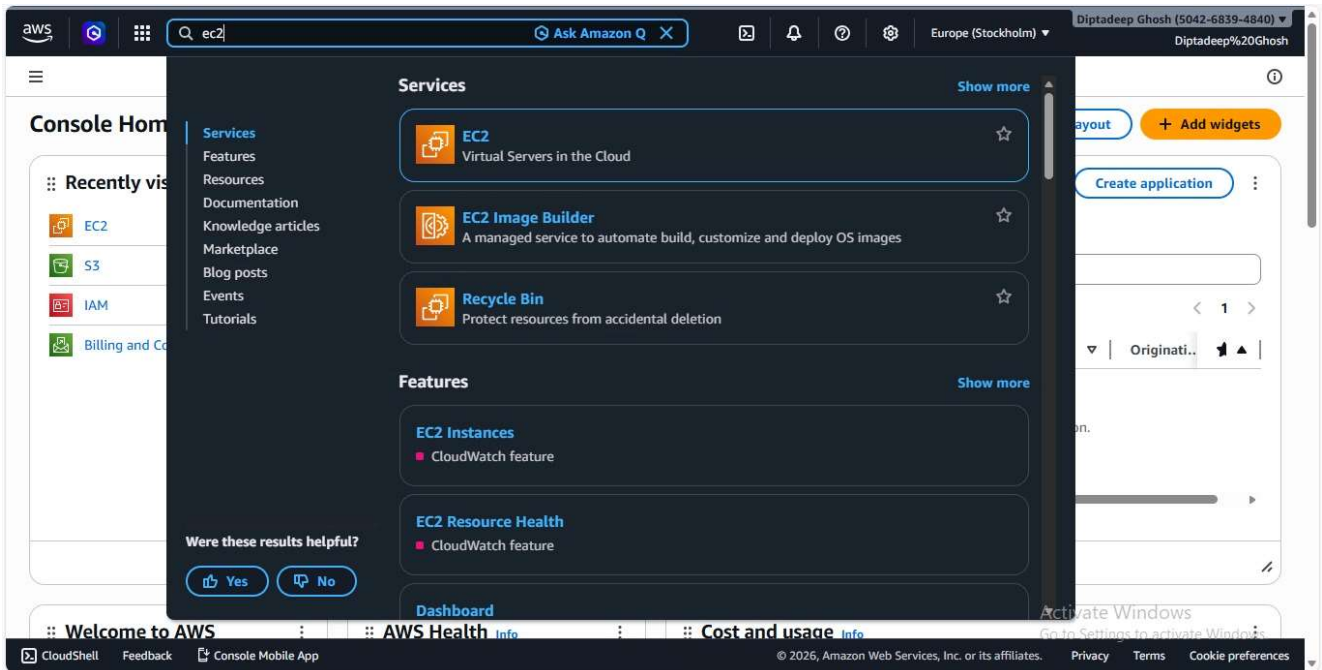
Solution:

Step 1: Launch an EC2 Instance

1. Login to AWS Management Console.

The screenshot shows the AWS Management Console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and the current region 'Europe (Stockholm)'. Below this, the 'Console Home' page is displayed. On the left, there's a 'Recently visited' section with links to EC2, S3, IAM, and Billing and Cost Management. On the right, there's an 'Applications (0)' section for the 'eu-north-1' region, showing a 'Create application' button. The bottom of the console features a 'Welcome to AWS' section, an 'AWS Health' section, and a 'Cost and usage' section. The footer includes links to CloudShell, Feedback, and the Console Mobile App, along with copyright information for Amazon Web Services, Inc.

2. Search for EC2 and open the service.



3. Click Launch instance.

Instance Configuration

4. Enter:

- Name: WebServer-EC2

5. Select:

- AMI: Ubuntu (Free Tier Eligible)

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name: [Add additional tags](#)

Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Summary

Number of instances Info: 1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.10...[read more](#)
ami-0317b0f0a0144b137

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

6. Choose:

Instance summary for i-03246ac9fe46bb0b5 Info

Updated 19 minutes ago

Instance ID
i-03246ac9fe46bb0b5

Public IPv4 address
65.2.29.1 | [open address](#)

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-47-53.ap-south-1.compute.internal

Instance type
t3.micro

VPC ID
vpc-0aa3b70c42e4d9b1c

IPV6 address
-

Hostname type
IP name: ip-172-31-47-53.ap-south-1.compute.internal

Answer private resource DNS name
IPv4 (A)

Auto-assigned IP address
65.2.29.1 [Public IP]

Actions

- Stop instance
- Start instance
- Reboot instance
- Hibernate instance
- Terminate (delete) instance

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
[Learn more](#)

- Instance type: t3.micro

7. Click Create new key pair:

- Name: webkey
- Type: RSA
- Format: .pem

Create key pair

Key pair name
Key pairs allow you to connect to your instance securely.

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

[Cancel](#) [Create key pair](#)

8. Download and save the key file securely.
9. Click on **Create security group** and check the boxes of **Allow HTTPS traffic from the internet** and **Allow HTTP traffic from the internet** and click on **Launch instance** button.

Network settings [Info](#) [Edit](#)

Network [Info](#)
vpc-0aa3b70c42e4d9b1c

Subnet [Info](#)
No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)
Enable

Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

- ☒ Allow SSH traffic from
Helps you connect to your instance
Anywhere
0.0.0.0/0
- ☒ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server
- ☒ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-019715e0d74f695be

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security

10. The Instance will be successfully created, return to the **Instance** page.

Success
Successfully initiated launch of instance (i-03246ac9fe46bb0b5)

Launch log

Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
[Create billing alerts](#)

Connect to your instance
Once your instance is running, log into it from your local computer.
[Connect to instance](#)
[Learn more](#)

Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
[Connect an RDS database](#)
[Create a new RDS database](#)
[Learn more](#)

Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots.
[Create EBS snapshot policy](#)

Activate Windows
Go to Settings to activate Windows

11. The Instance will be seen running, click on the newly created instance.

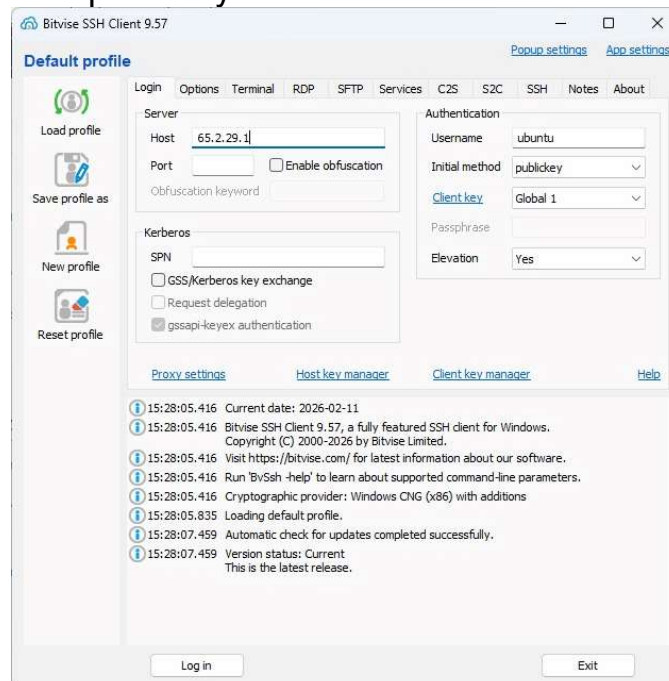
The screenshot shows the AWS Management Console interface for an EC2 instance. The left sidebar contains navigation links for EC2, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Capacity Manager, Images, AMIs, AMI Catalog, and Elastic Block Store. The main content area displays the 'Instance summary for i-03246ac9fe46bb0b5'. The instance is in the 'Running' state. The Public IPv4 address is 65.2.29.1. The instance type is t3.micro. The VPC ID is vpc-0aa3b70c42e4d9b1c. The Subnet ID is subnet-0c8be5b073be8e0e2. The instance is located in the Asia Pacific (Mumbai) region.

Step 2: Copy Public IPv4 Address

After the EC2 instance is launched and running, copy the Public IPv4 address from the Instance details page.

Step 3: Open Bitvise SSH Client and Configure Connection

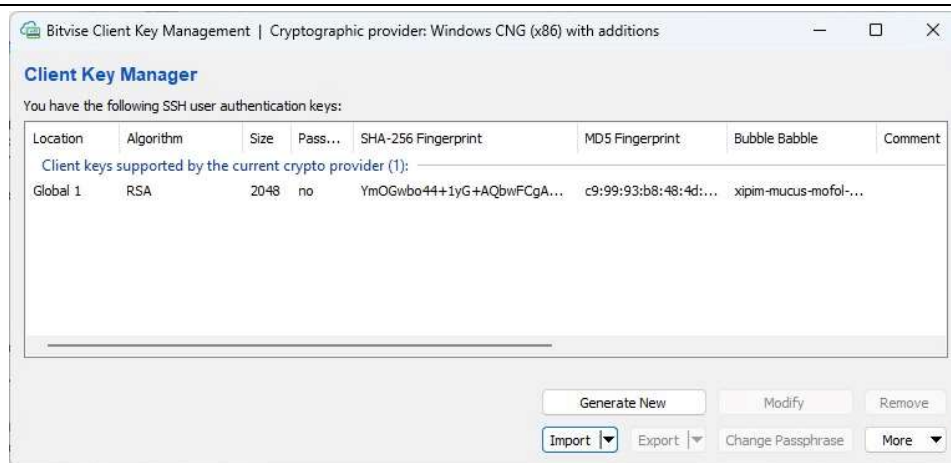
1. Open Bitvise SSH Client on the computer.
2. Paste the copied Public IPv4 address into the Host field.
3. Enter ubuntu in the Username field.
4. Set Initial method to publickey.



5. Click on Client key manager.

Step 4: Import Private Key

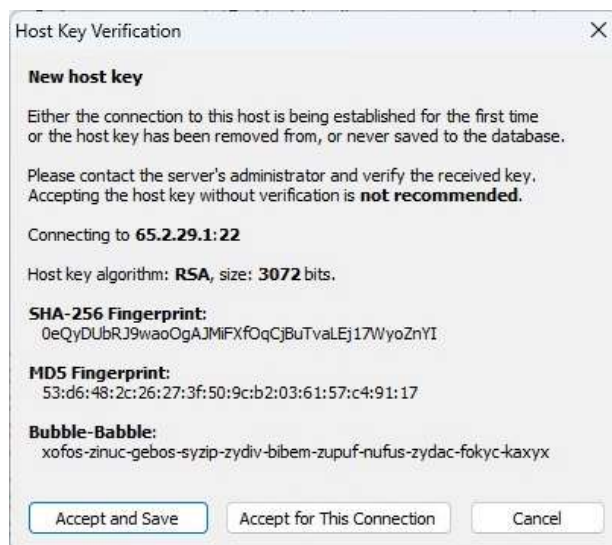
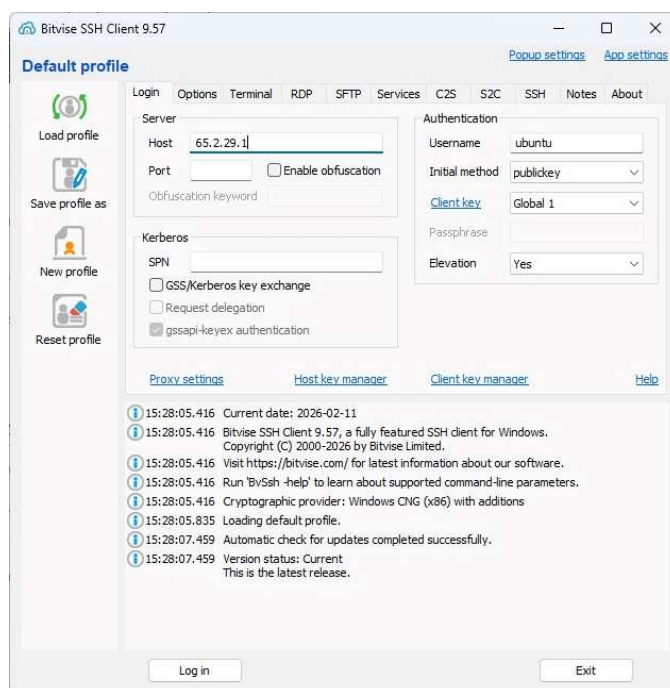
1. In Client Key Manager, click Import.
2. Select the downloaded .pem key file from the Desktop.



3. Save the key.

Step 5: Login to the Instance

1. Ensure Client key is set to Global 1.
2. Click on the Log in button.



Step 6: Open Terminal Console

After successful login:

```
ubuntu@65.2.29.122 - Bitwise xterm - ubuntu@ip-172-31-47-53: ~
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Wed Feb 11 10:08:29 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   25.9% of 6.71GB Processes:      111
Memory usage: 22%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 172.31.47.53

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-47-53:~$
```

- Click on **New Terminal Console** from the **Bitwise** side panel.

Step 7: Update Ubuntu System

Run the following command to update and upgrade the system:

sudo apt-get update && sudo apt-get upgrade -y

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

```
ubuntu@ip-172-31-47-53:~$ sudo apt-get update
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
```

```
ubuntu@ip-172-31-47-53:~$ sudo apt-get upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages will be upgraded:
 base-files bsdxtrautils bsdxutils dirmngr eject fdisk fwupd gir1.2-glib-2.0 gnupg gnupg-l10n
 gnupg-utils gpg gpg-agent gpg-wks-client gpgconf gpgsm gpgv inetutils-telnet initramfs-tools
 initramfs-tools-bin initramfs-tools-core keyboxd klibc-utils kpartx libblkid1 libc-bin libc6
 libdrm-common libdrm2 libfdisk1 libfwupd2 libglib2.0-0t64 libglib2.0-bin libglib2.0-data
 libklibc libldap-common libldap2 libmbim-glib4 libmbim-proxy libmbim-utils libmount1
 libnss-systemd libnuma1 libpam-systemd libpng16-16t64 libpython3.12-minimal libpython3.12-stdlib
 libpython3.12t64 libsmartcols1 libsodium23 libssl3t64 libsystemd-shared libsystemd0 libtasn1-6
 libudev1 libuuid1 libxml2 libxslt1.1 linux-tools-common locales motd-news-config mount
 multipath-tools numactl openssl python-apt-common python3-apt python3-distupgrade python3-pyasn1
 python3-urllib3 python3.12 python3.12-minimal screen snapd systemd systemd-dev systemd-resolved
 systemd-sysv telnet ubuntu-release-upgrader-core udev util-linux uuid-runtime
83 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Need to get 78.5 MB of archives.
After this operation, 776 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Step 8: Install Nginx Web Server

Install Nginx using the command:

sudo apt-get install nginx -y

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 564 kB of archives.
After this operation, 1596 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
```

Activate
Go to Settings

Step 9: Set Permissions for Web Directory

Navigate to the web directory:

cd /var/www

Give proper permissions to the HTML directory:

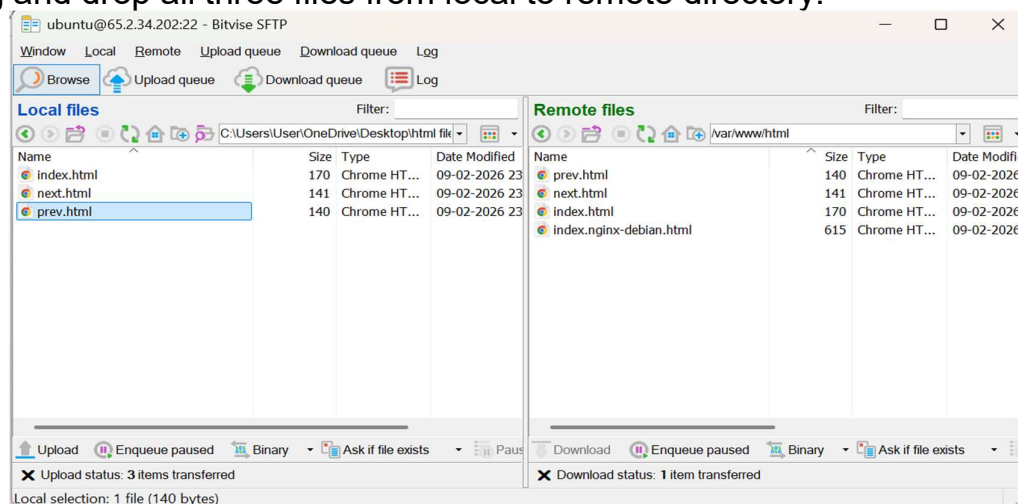
sudo chmod 777 html/

```
ubuntu@ip-172-31-47-53:/var$ cd www
ubuntu@ip-172-31-47-53:/var/www$ ls
html
ubuntu@ip-172-31-47-53:/var/www$ sudo chmod 777 html
ubuntu@ip-172-31-47-53:/var/www$
```

Activate
Go to Settings

Step 10: Upload Website Files Using SFTP

1. Click New SFTP window in Bitvise.
2. In the left pane (Local Files), navigate to the folder containing:
 - index.html
 - prev.html
 - next.html
3. In the right pane (Remote Files), navigate to: **/var/www/html**
4. Drag and drop all three files from local to remote directory.



Step 11: Access the Website

Open a new browser tab and enter:

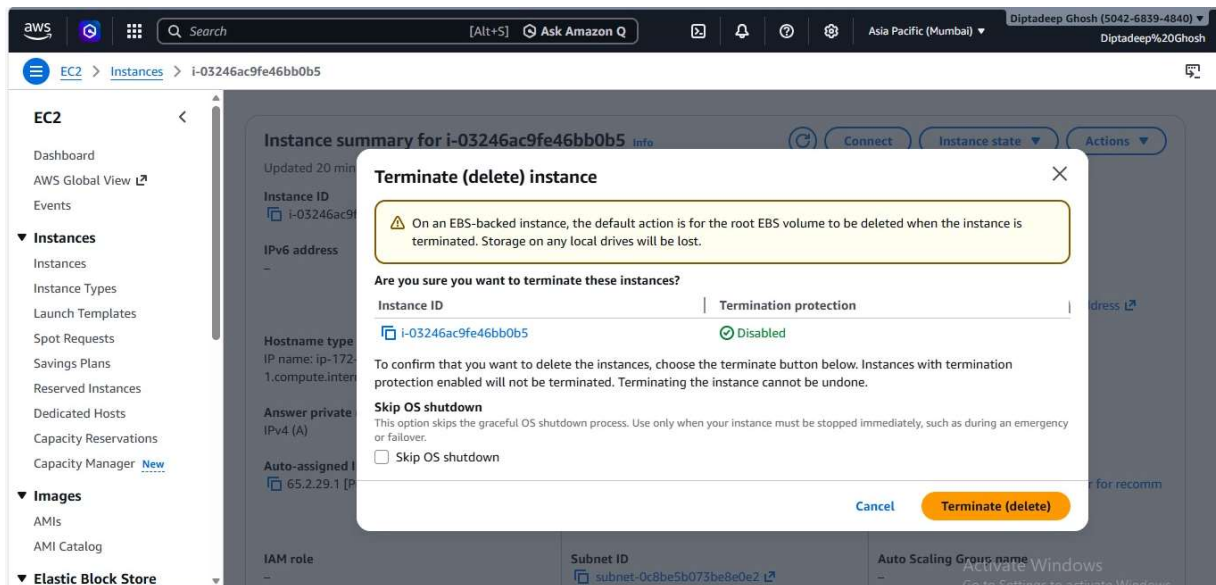
http://<Public-IPv4-address>

The static website will load successfully.



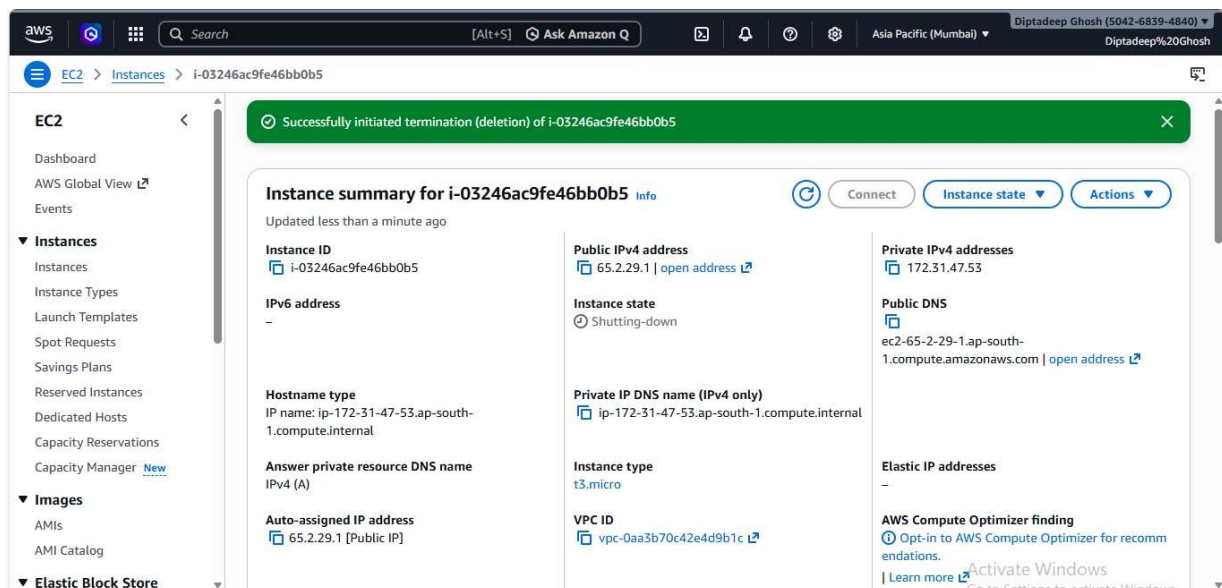
Step 13: Terminate the EC2 Instance

1. Select the same instance again.
2. Click Instance state → **Terminate (delete) instance.**
3. Confirm by clicking Terminate (delete).



Step 14: Instance Termination Confirmation

The EC2 instance is successfully terminated and all associated compute resources are released.



Conclusion

A static website consisting of multiple HTML files was successfully hosted on an Ubuntu-based EC2 instance using the Nginx web server and accessed via a public IPv4 address.